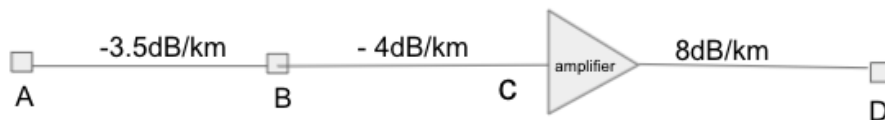


CSE320 (Data Comm) – Practice Sheet on Chapter 3

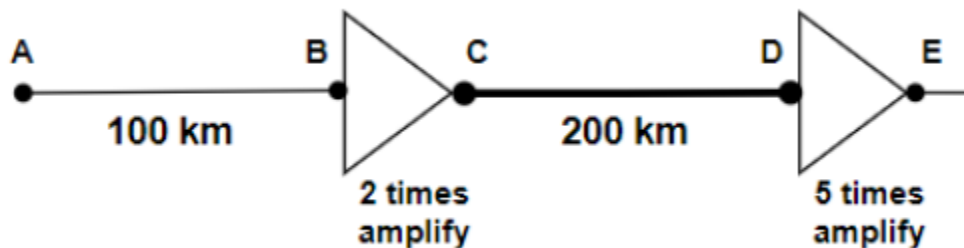
Q1.

Suppose $AB = 0.5\text{km}$, $BC = 0.6\text{km}$ and $CD = 0.5\text{km}$. The signal has a power of 3mW at the beginning of the cable (point A). **Predict** if the signal has gained or lost power at point D? (measure in dB)



Q2.

Suppose the signal power is 6 MW at point A. The power loss rate at the wire from A to B is 3 kW/km and from C to D is 0.04 dB/km . Calculate the total change of signal power in decibel and comment if the power is being amplified/attenuated.



Q3.

Consider a channel with a bandwidth of 2 MHz . The strength of the intended signal is 40mW and 60 times stronger than the noises present in the channel.

- I. **Interpret** the upper limit of the data rate that the channel can carry?
- II. In practice, better error performance can be achieved if we use a lower data rate. Assume we choose a data rate $\frac{2}{3}$ of the maximum upper limit. How many signal levels are needed to achieve this data rate?
- III. If we change the bandwidth to 4 MHz . How changes in signal levels should be made to maintain the same data rate?

Q4.

Consider a communication channel that needs to send 108 GB within 8 hours. The link operates on signals with a frequency range from 800 KHz to 13 MHz. If the link is perfect, i.e., no noise is introduced in the link,

- a. Determine the number of voltage levels needed to fulfill the requirement.
- b. In practice, there is no noise-free channel. Suppose, the strength of the noise power is 30mW which is 20 times weaker than the signal power. What will be the channel capacity considering the noise?

Q5.

If a periodic signal is decomposed into six sine waves with frequencies of 126, 348, 544, 896, 957 and 1900 Hz, what is its bandwidth? Draw the spectrum (frequency-domain representation), assuming all components have a maximum amplitude of 20 V.

Q6.

Suppose the lower frequency of a channel is 1863 MHz, and the higher frequency of the channel is 1931 MHz. If the upper limit of the bit rate is about 38.7 Mbps, what is the signal to noise ratio in dB?

Q7

A periodic composite signal is characterized by a bandwidth of 20 kHz and a starting (minimum) frequency of 25 kHz. The signal exhibits peak amplitudes of 12 V, 18 V, and 24 V corresponding to the lowest, highest, and midpoint frequencies within its range, respectively. Calculate the maximum frequency. Draw the frequency domain plot.

Q8

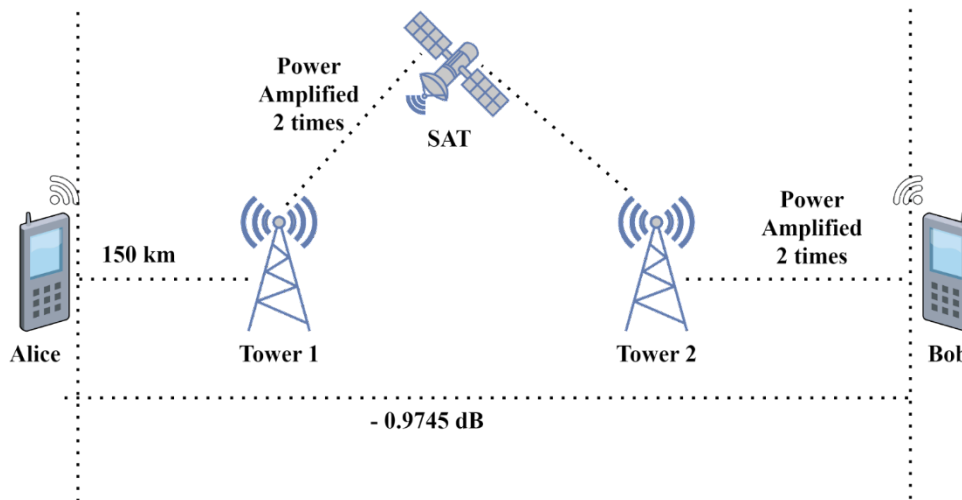
A document takes 0.5 minutes to download over a communication channel, assuming the channel is completely noiseless. The data rate of the channel is 10 Mbps and the bandwidth is 1.5 MHz, analyze the following aspects:

- i. Calculate the number of characters present in the document.
- ii. Determine the value of the signal level.

Q9

Alice is calling Bob through a hybrid communication system as shown in the following figure. Suppose the signal power is 801 mW at Alice's phone. The power loss rate from Alice to Tower 1 is 2.14 mW/km. The final attenuation from Alice to Bob is -0.9745 dB.

- Calculate the attenuation for the signal transmitted from SAT to Tower 2.



Q10

Consider a communications channel being used by a cable modem network. The channel has use of the spectrum between 104MHz and 119MHz. The signal power is 22mW and the noise power is 2mW.

- Interpret the theoretical maximum capacity of the channel in bps.
- Assuming the capacity of the channel could be realized, determine how many signal levels would be needed.
- Discuss what advantage/disadvantage, if any, would there be in using twice this number of signal levels.